- Sperms are motile.
- Flower like structures are absent. Strobili are simple.
  Example: *Cycas, Zamia*.

Class II – Coniferopsida

- Tall trees with simple leaves of varied shape.
- Wood is pycnoxylic.
- Cone like strobili are present.
- Motile sperms are absent (except *Ginkgo biloba*). Example: *Pinus*.

Class III – Gnetopsida

- Shrubs, trees and lianas.
- Leaves are elliptical or strap-shaped, simple, opposite or whorled.
- Motile sperms are absent.
- Wood contains vessels.
- Strobili is called as inflorescence.
- Flower like structure with perianth is present. Example: *Gnetum, Ephedra*.

2.6.3 Comparison of Gymnosperm with Angiosperms

Gymnosperms resemble with angiosperms in the following features

- Presence of well organised plant body which is differentiated into roots, stem and leaves.
- Presence of cambium in gymnosperms as in dicotyledons.
- Flowers in *Gnetum* resemble the male flower of the angiosperm.
- The zygote represent the first cell of sporophyte.
- Presence of integument around the ovule.
- Both plant groups produce seeds.
- Pollen tube helps in the transfer of male nucleus in both (siphonogamy).
- Presence of eustele.

The difference between Gymnosperms and Angiosperms were given in Table 2.5

Table 2.5: Difference between Gymnosperms and Angiosperms

S.No	Gymnosperms	Angiosperms
1.	Vessels are absent [except Gnetales]	Vessels are present
2.	Phloem lacks companion cells	Companion cells are present
3.	Ovules are naked	Ovules are enclosed within the ovary
4.	Wind pollination only	Insects, wind, water, animals etc., act as pollinating agents
5.	Double fertilization is absent	Double fertilization is present
6.	Endosperm is haploid	Endosperm is triploid
7.	Fruit formation is absent	Fruit formation is present

2.6.4 Economic importance of Gymnosperms

Table 2.6: Economic importance of Gymnosperms

S.No	Plants	Products	uses
1.	*Cycas circinalis, Cycas revoluta*	Sago	Starch used as food
2.	*Pinus gerardiana*	Roasted seed	Used as a food
3.	*Abies balsamea*	Resin (Canada balsam)	Used as mounting medium in permanent slide preparation
4.	*Pinus insularis, Pinus roxburghii*	Rosin and Turpentine	Paper sizing and varnishes
5.	*Araucaria* (Monkey's puzzle), *Picea* and *Phyllocladus*	Tannins	Bark yield tannins and is used in Leather industries
6.	*Taxus brevifolia*	Taxol	Drug used for cancer treatment

11th Botany_EM_Chapter 2.indd 52 06/01/2023 19:41:33